

JEREMIAH EMRICH

Data Analyst — Python / SQL

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SUMMARY

Data analyst building interactive dashboards and applied AI tooling with Python, SQL, Pandas, and Power BI — from raw data through deployed, production insight tools. Shipped a live dashboard analyzing 39,000+ federal aviation safety records, an end-to-end Python → SQL → Power BI analysis of Kansas property tax trends built from raw government PDF sources, and a full-cycle personal finance analytics tool (data cleaning, SQL analysis, predictive modeling, forecasting). Also builds AI-augmented analysis tooling — retrieval-augmented generation pipelines and LangGraph agents — combining traditional analytics with applied AI, an increasingly expected skill set as the field shifts toward AI-augmented analysis. FAA-certified Airframe mechanic background adds regulatory and safety-critical domain expertise. Completing a B.S. in Computer Science.

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript, JavaScript

Data & Analysis: Pandas, NumPy, Excel, Power BI, Streamlit, Plotly, Recharts, Statistical Analysis, Data Visualization, Scikit-learn

Databases: PostgreSQL, SQLite, Redis, Schema Design, Data Modeling, Query Optimization

AI / LLM Tooling: OpenAI API, Retrieval-Augmented Generation (RAG), ChromaDB, Vector Embeddings, LangGraph, LangChain

Tools & DevOps: Git, GitHub Actions, Docker, Railway, Jupyter Notebook, Linux

Web (supporting): Django, Django REST Framework, React, HTML, CSS

PROJECTS

Kansas Property Tax Analysis — Python, Pandas, SQL, **Power BI**

[GitHub](#)

- Built an end-to-end Python → SQL → Power BI pipeline answering a real civic/personal question: is Kansas property tax growth driven by rising valuations or rising mill levy rates, and would relocating to a neighboring state actually lower the bill?
- Extracted and validated data from unstructured government PDFs (KDOR mill levy tables, statewide historic tax records, Lincoln Institute 50-State Property Tax Comparison Study) using pdfplumber with content-based table location, loading results into a SQLite database with SQL views feeding the dashboard
- Found that Reno County's mill levy rate fell 5.6% (2020–2025) while statewide assessed valuation grew 36%, isolating valuation growth — not rate increases — as the actual driver; benchmarked Wichita against Kansas City, Omaha, Oklahoma City, and Denver, showing Wichita held the lowest median tax bill of the five in every year 2023–2025

FAA Aviation Incident Analysis Dashboard — Python, Pandas, **Streamlit**, **Plotly**

[Live Demo](#)

- Built an interactive dashboard analyzing 39,560 NTSB aviation incident records (2000–present) with trend lines, injury-level breakdowns, weather-condition analysis, and a US choropleth heatmap of incident density by state
- Grounded every insight in FAA A&P domain expertise — interpreting patterns (e.g., VMC vs. IMC incident rates, aircraft-type frequency vs. risk) with real aerospace regulatory context, not just raw statistics

Personal Finance Analysis Dashboard — Python, Pandas, SQL, **Streamlit**, **Plotly**, Scikit-learn

[Live Demo](#)

- Built an end-to-end analytics pipeline across 1,500 transactions (2020–2024): synthetic data generation, data cleaning, SQL-based analysis in SQLite, interactive Plotly visualizations, and a 6-month linear regression forecast
- Documented the full workflow across 4 Jupyter notebooks (cleaning, SQL queries, visualization, forecasting) with dynamic KPI filtering and income/expense breakdowns by category and day of week

Fit For My Role — Python, Django, OpenAI API, **ChromaDB**

[fitformyrole.com](#)

- Built a retrieval-augmented generation pipeline (pdfplumber → ChromaDB vector store → similarity search) to ground AI-generated resume/job match scoring in real data, integrating live job-board APIs (Remotive, Arbeitnow) for on-demand scoring

AI Hunting Chat — Python, **LangGraph**, **LangChain**, OpenAI

[aihuntingchat.com](#)

- Built an AI agent that parses plain-English requests, selects the correct tool (live weather data, calculation, or an internal knowledge base), executes it, and responds in natural language with cross-turn conversation memory

EDUCATION & CERTIFICATIONS

B.S., Computer Science — University of Maryland Global Campus

Expected 2029 (72/120 credits)

Relevant coursework: Algorithms, Data Structures, Databases, Software Engineering

- Codecademy — Career Path: Business Intelligence Data Analyst (SQL, Python, Pandas, data visualization, statistical analysis, BI reporting)
- Codecademy — Career Path: Machine Learning/AI Engineer (ML fundamentals, AI model integration, production AI tooling)
- Codecademy — 13+ individual course certifications, including Learn Python 3, Learn SQL, Learn Intermediate Python 3: OOP

A.A.S., FAA Airframe Mechanic — WSU Tech

2019 – 2021

FAA Airframe & Powerplant (A&P) Certificate — safety-critical systems, technical documentation, precision diagnostics